

1) Data Mining:

* It is the process of extracting useful, hidden and meaningful patterns (or) knowledge from large amount of data using statistical, machine learning and database techniques.

Two basic data mining tasks:

classification - Assign data into pre-defined categories.

clustering - Groups similar data objects without pre-defined labels.

KDD process:

- * data cleaning.
- * data integration.
- * data selection.
- * data transformation.
- * data mining.
- * pattern evaluation.
- * knowledge presentation.

Role of data processing:

- * improves missing and noisy data.
- * improves data quality.
- * increases the accuracy and efficiency

of data mining.

b) data cleaning:

* It is the process of removing errors, missing values, and inconsistent data from a dataset.

Two data quality issues:

- * missing values.
- * noisy data.

Handling missing values:

- * replace with mean / median / mode.
- * delete records with many missing values.

Handling noisy data:

- * use binning.
- * remove outliers using clustering (or) smoothing.

2.) data integration:

* It is the process of combining data from different sources into one consistent dataset.

challenges:

- * data redundancy.
- * data inconsistency.

Handling:

Redundancy: Remove duplicate data.

Inconsistency: standardize formats and resolve conflicts.

b.) covariance analysis:

* covariance analysis measures how two variables change together.

Two covariance formulas:

i.) population covariance.

$$\text{cov}(x, y) = \frac{\sum (x - \mu_x)(y - \mu_y)}{N}$$

ii.) sample covariance:

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

3.) Given data :

stock A

stock B

2

5

3

8

5

10

4

11

6

14

Analysis:

As stock A increases, stock B also generally increases.

Conclusion:

* they move together.

* they have positive covariance.

b.) Architecture of data mining system.

* data source.

* database / data warehouse.

* data cleaning & integration.

* data mining engine.

* pattern evaluation.

* Knowledge base.

* User interface.

Role :-

these components work together to collect, process, analyze and present user knowledge.

4) Given :

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

mean :

$$\bar{x} = \frac{4+5+6+7+9}{5}$$

$$= 6.2$$

$$\bar{y} = \frac{8+3+7+5+6}{5}$$

$$= 5.8$$

sample covariance :

$$\text{cov} = \frac{-1.80}{4}$$

$$= -0.45$$

b) Data preprocessing steps :

- * data cleaning.
- * data integration.
- * data transformation.
- * data reduction.
- * data discretization.

pipe line :

Raw data $\rightarrow$ cleaning $\rightarrow$ missing value handling $\rightarrow$ outlier removal $\rightarrow$ normalization $\rightarrow$ ready data.

5.) equal - frequency binning:

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22 $\rightarrow$ mean = 18.

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35 $\rightarrow$ mean = 28.11.

Bin 3: 35, 35, 35, 36, 40, 43, 46, 52, 70 $\rightarrow$ mean = 43.78.

* this mean values replace the original values for smoothing.

b.) data discretization.

* It is the process of converting raw data into intervals.

importance :

- * reduces data complexity.
- * removes noise.
- * improves mining efficiency.

comparison :

equal - width

equal interval size.

Equal - frequency.

equal no. of records.